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Dr. Martijn van den Heuvel
Editor, Human Brain Mapping

May 5, 2025

Dear Prof. van den Heuvel,

Last July, we submitted our original manuscript HBM-24-0780, entitled, “*Enhanced Detection of Age-Related and Cognitive Declines Using Automated Hippocampal-to-Ventricle Ratio in Alzheimer's Patients*” for publication in Human Brain Mapping as suggested by both Louis Lemieux and Nathan Sprang (editorial board member and associate editor, respectively). After a first round of review from an expert reviewer and an associate editor, in an email Nov. 11, 2024 you offered that we could revise and resubmit. In this resubmission, we include two versions of the updated manuscript—one clean version with no change tracking and another with the tracking of the changes—as well as a point-by-point response to all of the reviewer’s and editor’s comments. As suggested by the associate editor, we have included an additional out of sample validation set, include a link to our open software with a container implementing the full segmentation pipeline as well as the scripts to compute the HVR.

As indicated in our initial submission:

- hippocampal atrophy is an early and important sign of neurodegeneration in Alzheimer’s disease;
- manual segmentation of hippocampus is difficult and time consuming;
- we have automated the Hippocampal-to-Ventricle Ratio (HVR) as a more robust biomarker compared to hippocampal volume alone for quantification of medial temporal lobe neurodegeneration;
- we compared Multi-Atlas Label Fusion (MALF), Nonlinear Patch-Based Segmentation (NLPB), and a Convolutional Neural Network (CNN) as well FreeSurfer for segmentation.
- we evaluated on large cohort of participants (n=1641) from ADNI.
- Key findings include the superior performance of the CNN-based segmentation method in terms of accuracy, reliability, and clinical relevance compared to MALF and NLPB.
- HVR yielded higher control:AD effect sizes than HC volumes among all segmentation methods

We stand by our belief that this work will be of interest to the broad audience of Human Brain Mapping. Our results give insight into the importance of high-quality segmentation of the hippocampus for the study of neurodegeneration, and our results can help users decide which is the best method to use for their application.

We certify that this manuscript is still submitted only to Human Brain Mapping, and it has not been, nor will be published in or submitted to any other journal. Please note that a very early preprint version of the work is available online through bioRxiv (BIORXIV/2024/588928). We have no affiliation, financial agreement or any other involvement with any company in connection with this research study. All authors have read the manuscript, have full access to the data and the right to publish with the data and take full responsibility for the data, analyses, interpretation and the conduct of research. On the following page, we include the same list of suggested reviewers as included previously.

Thank you for considering again our work for publication in Human Brain Mapping.

Sincerely,

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List of suggested reviewers:

- Josephine Barnes, University College London, Dementia Research Centre (j.barnes@ucl.ac.uk)
- Marie Chupin, ARAMIS team of the Brain and Spine Institute (marie.chupin@cnrs.fr)
- Paul Aljabar, Imperial College London (paul.aljabar@imperial.ac.uk)
- Marina Boccardi, Department of Psychiatry, LANVIE-Laboratory of Neuroimaging of Aging, University of Geneva, (mboccardifbf@googlemail.com)
- Eleanor Maguire, Wellcome Centre for Human Neuroimaging, UCL, (e.maguire@ucl.ac.uk)
- Ali Khan, assistant professor, University of Western Ontario (ali.khan@uwo.ca)
- Sean Nestor, assistant professor, Neurosciences and Clinical Translation, Sunnybrook, (sean.nestor@sunnybrook.ca)
- Bruce Fischl, MGH, Harvard, (bfischl@mgh.harvard.edu)
- Hongzhi Wang, IBM Almaden Research Center (hongzhiw@mail.med.upenn.edu)